

Total No. of Questions : 8]

SEAT No. :

PE2736

[Total No. of Pages : 2

[6583]-317

T.E. (Cyber Security) (Honors)

ENTERPRISE ARCHITECTURE AND COMPONENTS

(2019 Pattern) (Semester - VI) (310403)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicates full mark
- 3) Draw neat diagram wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Describe how a component relationship diagram aids in system design. [5]
b) Define a component model in software engineering. [6]
c) How would you use a component description to document a software component? Provide an example. [6]

OR

- Q2)** a) Describe how component relationship diagrams help in understanding system architecture. Provide an example to support your explanation. [9]
b) Design a component interaction diagram for a real-time messaging application, and explain your design choices. [8]

- Q3)** a) Define 'operational model' in the context of software engineering. [6]
b) Describe the role of service qualities in shaping operational model design decisions. [6]
c) List three service qualities relevant to operational models. [6]

OR

- Q4)** a) Explain the role of context in selecting operational model design techniques for a given project. Provide examples to support your explanation. [9]
b) Discuss the advantages and disadvantages of various operational model design techniques in different project contexts. Use specific scenarios to illustrate your points. [9]

P.T.O.

- Q5) a)** Identify and discuss service qualities crucial for effective metadata management. Explain how these qualities contribute to the success of metadata management initiatives. [6]
- b)** Define “metadata” and “master data” in the context of data management. Explain the importance of clear terminology in data-related projects. [6]
- c)** Provide definitions for ‘business scenario’ and ‘component deep dive’ in the context of software engineering. Explain how these terms are relevant to system design and development. [5]

OR

- Q6) a)** Identify and explain three service qualities relevant to metadata management systems. Discuss their importance in ensuring the effectiveness and reliability of such systems. [8]
- b)** Describe a business scenario where effective metadata management is critical. Explain how proper metadata management practices can benefit the scenario. [9]
- Q7) a)** Describe how SABSA, COBIT, and TOGAF frameworks can be used to develop an enterprise security architecture. Provide a step-by-step approach, highlighting the key considerations at each stage. [9]
- b)** Provide a real-life example of an organization that has successfully implemented SABSA, COBIT, and TOGAF frameworks in its enterprise architecture. Discuss the benefits achieved and challenges encountered during the implementation process. [9]

OR

- Q8) a)** Explain how CMMI can be used to monitor, measure, and report the progress of architecture development activities within an organization. Discuss the key process areas and metrics relevant to this task. [9]
- b)** Outline the steps involved in using SABSA, COBIT, and TOGAF to develop an enterprise security architecture. Discuss the specific contributions of each framework in the process. [9]

✕

✕

✕